COURSE INFORMATION

Faculty:	Computing	**Page:**	1 of 7
Program name:	Bachelor of Computer Science		
Course code:	SECV2113	**Academic Session/Semester:**	20232024/2
Course name:	Human-Computer Interaction	**Pre/co requisite (course name and code, if applicable):**	Nil
Credit hours:	3		

Course synopsis	This course will introduce students to human-computer interaction theories and design processes. The emphasis will be on applied user experience (UX) design. The course will present an iterative evaluation centred UX lifecycle and will introduce a broader notion of user experience, including usability, usefulness, and emotional impact. The lifecycle should be viewed as template intended to be instantiated in many ways to match the constraints of a particular development project. The UX lifecycle activities cover contextual inquiry and analysis, requirements extraction, design-informing models, design thinking, ideation, sketching, conceptual design, and formative evaluation.			
Course coordinator	Dr Nur Zuraifah Syazrah binti Othman			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail @utm.my**
	(01) Dr Izyan Izzati binti Kamsani			izyanizzati
	(02) Dr Nur Zuraifah Syazrah binti Othman			zuraifah

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	CLO	PLO (Code)	*Taxonomies and **generic skills	T&L methods	***Assessment methods
CLO1	Apply the concept of human–computer interaction and the application of human factors as a multi-level process of communication through UX design and evaluation of interactive systems.	PLO1 (KW)	C3	Lecture, active learning	A, F

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CLO2	Apply the theoretical knowledge to issues that arise in the design of interactive systems.	PLO3 (PS)	C3	Lecture, Project-based learning	GR, F
CLO3	Build a prototype development project according to principles and guidelines of human–computer interaction.	PLO5 (TH)	P3 TH1, TH3	Project-based learning	GR
CLO4	Ability to act responsibly in carrying out tasks throughout the project life cycle	PLO9 (GC)	A6 GC4	Project-based learning	PA
Refer *Taxonomies of Learning and **UTM's Graduate Attributes, where applicable for measurement of outcomes achievement *** L – Lecture; T – Tutorial; TE – Test; PS – Problem Solving; PR – Project; F – Final Exam					

Details on Innovative T&L practices:

No.	Type	Implementation
1.	Active learning	Conducted through in-class activities
2.	Project-based learning	Conducted through building a prototype. Tasks are given in sequential steps throughout the semester. Students in a group are asked to build a prototype to solve a challenge faced by users that require HCI and UX knowledge to deliver good apps or systems. The report must comply with the project specifications and be given in the form of a written report.

Weekly Schedule:

Week 1 17 Mar - 21 Mar 2024	Introductory to Course Introduction to Human Computer Interaction ● Good and poor design ● Interaction design in comparison to human-computer interaction ● User experience, accessibility, and inclusiveness ● Usability ● General design principles
Week 2 24 Mar – 28 Mar 2024	Section A: Human Behaviour Users: cognition aspects, emotional and social interactions ● Cognition aspects ● Emotional interactions * Assignment 1

Week 3 31 Mar - 4 Apr 2024	Users: cognition aspects, emotional and social interactions (cont.) ● Social mechanism
Week 4 7 Apr - 11 Apr 2024 *Hari Raya Aidilfitri (10, 11 Apr)	Conceptualising Interaction Design ● Understanding the problem space ● Conceptualising the design: ○ conceptual models ○ metaphors and analogies ○ interaction types
Week 5 14 Apr - 18 Apr 2024	Trends in Interfaces ● Interface types
Week 6 21 Apr - 25 Apr 2024	<u>Section B: Design</u> The Process of Interaction Design ● Involving users in development ● Principles of user-centred approach ● Four basic activities in interaction design ● AgileUX ● Practical issues * Project part 1
Week 7 28 Apr - 2 May 2024 *Labour Day (1 May)	Establishing Requirements ● Different kinds of requirements: persona, goals, environments ● Data gathering for requirements: interviews, focus groups, questionnaires, ethnography, diaries, contextual inquiry, studying documentation, researching similar products. ● Data gathering guidelines. ● Data analysis, interpretation, and presentation * Peer Assessment for Project 1 * Project part 2-a
Week 8 5 May – 9 May 2024	**MID SEMESTER BREAK**

Week 9 12 May – 16 May 2024	Establishing Requirements (cont.) ● Task description: scenarios, use case etc. ● Task analysis: HTA * Project part 2-b
Week 10 19 May - 23 May 2024 *Wesak Day (22 May)	<u>Section C: Implementation</u> Design, Prototyping and Construction ● Prototyping and construction ● Conceptual design ● Concrete design ● Scenarios in design ● Prototypes in design (storyboard, card-based) ● Construction: physical computing, SDK * Peer Assessment for Project 2
Week 11 26 May - 30 May 2024	Design, Prototyping and Construction (cont.) ● Gestalt Principle * Project part 3
Week 12 2 Jun - 6 Jun 2024	<u>Section D: Evaluation</u> Introducing Evaluation ● Why, what, where and when of evaluation ● Evaluation approaches and methods ● Inspection (heuristic evaluation, walkthrough) ● Analytics * Project part 4a (build prototype v1) * Prototype v1 Evaluation (Assignment 2)
Week 13 9 Jun - 13 Jun 2024	Other Evaluations Types ● Evaluation case studies ● How to: usability test, experiment, field studies * Peer Assessment for Project 3 * Prototype v2 Evaluation – test with users (Project part 4b)

Week 14 16 Jun - 20 Jun 2024 *Hari Raya Haji (17 Jun)	Data Analysis, Interpretation and Presentation ● Simple quantitative analysis ● Simple qualitative analysis ● Tools to support data analysis
Week 15 23 Jun - 27 Jun 2024	Project Presentation * Peer Assessment for Project 4
30 Jun - 4 Jul 2024	REVISION WEEK

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

Applying UX design practice, building prototyping (low and medium fidelity), performing usability testing, teamworking

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline	Teaching and Learning Activities						TOTAL SLT
	Guided Learning (Face to Face)				Guided Learning Non-Face to Face	Independent Learning Non-Face to face	
CLO	**L**	**T**	**P**	**O**			
CLO 1	23h	4h				8h	**35h**
CLO 2	4h		5h		5h	44h	**58h**
CLO 3	1h			2h	7h	8h	**18h**
CLO 4				3h			**3h**
Total SLT	**28h**	**4h**	**5h**	**5h**	**12h**	**60h**	**114h**

Continuous Assessment		PLO	Percentage	Total SLT
1	Assignment	PLO1	5	As in CLO 1
2	Project: Part 1	PLO3	5	As in CLO 2
3	Project: Part 2	PLO3	10	As in CLO 2
4	Project: Part 3	PLO3	10	As in CLO 2
5	Project: Part 4	PLO3, PLO5	20	As in CLO 2 & CLO 3
6	Mid Semester Assessment	PLO1	15	As in CLO 1
Final Assessment			Percentage	Total SLT
1	Peer Assessments	PLO9	5	3
2	Final Exam	PLO1, PLO3	30	3
Grand Total			**100**	**120h**

L: Lecture, T: Tutorial, P: Practical, O: Others

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

N/A

Learning resources:

Main references:

i) Sharp, H., Rogers, Y., Preece, J., (2019) "Interaction Design: Beyond Human-Computer Interaction", 5th Edition, New York: J. Wiley & Sons.

Additional references:

i) Baxter, Courage & Caine (2015), "Understanding Your Users"
ii) MacKenzie, I.S. (2013), "Human-Computer Interaction: An Empirical Research Perspective", Morgan Kauffman.
iii) Dix, A. et. al (2004), "Human-Computer Interaction", 3rd Edition, Prentice Hall.

Online

http://elearning.utm.my

Norman Group website https://www.nngroup.com/
Usability.gov website https://www.usability.gov
Interaction Design Foundation https://www.interaction-design.org

Academic honesty and plagiarism:

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and disciplinary steps may be taken by the faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

No	Assessment	PLO1	PLO3	PLO5	PLO9	TOTAL
		CLO1	CLO2	CLO3	CLO4	
1	ASSIGNMENT	20				20
2	PROJECT 1		5			5
3	PROJECT 2		10			10
4	PROJECT 3		10			10
5	PROJECT 4		5	15		20
6	PEER ASSESSMENT (4)				5	5
7	FINAL EXAM	20	10			30
TOTAL PLO		40	40	15	5	100